

# **Probing Human Event Perception through Large Language Models with Incremental Word Processing**

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## Abstract

Humans break continuous experiences into separate events to better understand the world, remember details, and respond to changes. Event boundaries happen when there are sudden shifts in context, like changes in actions or goals. Event Segmentation Theory (EST) explains that these boundaries are created when predictions about what comes next fail. Previous studies have used computational methods such as large language models (LLMs) to predict event boundaries in narratives, and their findings showed strong alignments with how humans segment events. Yet, how these alignments imply about human perceptions about events is poorly understood. In our research, we aim to investigate the event perception of humans through LLMs. We used multiple LLMs to directly probe the event boundary with varying context lengths. By incrementally providing the contexts, the models need to answer ‘Yes or No’ to decide whether the current word is a boundary between events. Generally, a larger model aligns better with humans’ judgement density than smaller models. Moreover, shorter context lengths of the same LLM align better than longer context lengths with human judgement density. This result indicates that local attention on the story context in LLM is more human-like than the global attention, which potentially mimics human memory mechanism in this task. In summary, our project uses LLMs to predict and interpret human event segmentation, shedding lights on the intersection of cognitive sciences and AI.

**Keywords:** Event Segmentation, Large Language Models (LLMs), Cognitive Science, AI Alignment

## Introduction

Human cognition tends to break continuous experiences into distinct events. These event boundaries mark key changes in experience and are very important for understanding stories, aiding memory and promoting decision-making. The Event Segmentation Theory (EST), developed by Zacks et al. (2007), suggests that event boundaries arise when there are prediction errors—moments when the expectations about what happens next are violated. These prediction errors force an update of the mental model of the ongoing event, signaling a transition to a new event.

In recent years, computational models, particularly large language models (LLMs), have been explored for their potential to replicate behaviors (Binz & Schulz, 2023). In event segmentation, LLMs, such as GPT-3 and GPT-4, have demonstrated the ability to segment narrative events in a manner that is highly congruent with human segmentation. Prior work has shown that LLMs can successfully identify boundaries between events in natural language text, using techniques such as Bayesian surprise and embedding shifts to quantify prediction errors (Kumar et al., 2023; Michelmann et al., 2023). These studies suggest that LLMs are not only effective in processing linguistic data but also provide insights into the underlying cognitive mechanisms that govern event perception.

However, despite these promising findings, the extent to which LLMs can fully replicate the complexities of human event segmentation remains poorly understood. While LLMs show great consistency with human judgment, the nature of this consistency and its implications for our understanding of human cognition are still an open question. This research seeks to bridge this gap by directly probing how LLMs handle event boundaries in relation to human event perception,

using various model types and context lengths to examine the consistency between computational predictions and human segmentation data.

## **Related Work**

Event Segmentation Theory (EST) helps explain how humans break continuous experiences into separate events. According to EST, event boundaries appear when there's a large gap between what we expect to happen and what actually happens. These gaps, called prediction errors, prompt us to update our mental model of the situation, signaling a shift to a new event. This theory is supported by research showing that humans divide their experiences based on prediction errors. Events marked by these boundaries are easier to remember, and people recall information from event boundaries better than information within an event. This ability to segment experiences is not just important for understanding stories but also plays an indispensable role in activities like action performance and processing time, showing how it is central to cognition.

In recent years, researchers have investigated whether large language models (LLMs) can replicate human event segmentation. Models like GPT-3 have shown promise in this area. In 2023, Kumar et al. showed that LLMs like GPT-2 and GPT-3 can identify event boundaries using a metric called Bayesian surprise. This metric can measure prediction errors in the model's output and link these errors to human-defined event boundaries. This approach mirrors the principles of EST, suggesting that both humans and LLMs may detect event boundaries using similar mechanisms.

Michelman et al. (2023) expanded on this idea by showing that GPT-3 does not just segment text, but does so in a way that matches human judgments. The boundaries GPT-3 identified were strongly correlated with those marked by human annotators. This suggests that LLMs can detect

event boundaries in a way like how humans do. These findings point to the potential for LLMs to help study human event perception and offer a way to automate event segmentation in long texts.

Bayesian surprise is key to this process. It tracks how much the model’s predictions about the next words in a text are off. When the error spikes, signaling something unexpected, the model recognizes this as an event boundary. Studies have shown that these spikes in Bayesian surprise match human judgments of event boundaries, supporting the idea that both humans and LLMs use similar cues to detect transitions between events.

Despite these promising results, some questions still remain. While LLMs have shown they can be consistent with human event segmentation in certain cases, we do not yet fully understand how this works across different kinds of narratives, models, and settings. For example, how do factors like model size, context length and word frequency impact the accuracy of event boundary predictions? These factors may be important and need more research.

## **Research Questions**

This study is motivated by the desire to better understand the consistency between human event segmentation and LLM predictions. Specifically, we address the following research questions:

1. Can LLMs effectively replicate human event boundary judgments across different narrative contexts in an incremental processing setup?
2. How do different models and context lengths influence alignment with human segmentation performance?
3. Does incorporating prior distributions of word frequency improve model alignment with human judgments?

The central aim of this research is to assess whether LLMs can accurately replicate human event boundary judgments across varying narrative contexts and to determine how model type, context

length, and word frequency distributions influence the alignment between computational models and human event segmentation performance.

## **Methods**

### **Materials**

This study utilized three narrative texts: *Monkey in the Middle* (30 minutes), *Pieman* (7.5 minutes), and *Tunnel Under the World* (25 minutes). These stories were selected for their diversity in terms of narrative length and complexity, providing a broad basis for analyzing event segmentation. Human event boundaries were obtained from previous studies and aggregated using Gaussian smoothing (Kumar et.al., 2023), creating a continuous density function that represents human judgment for event transitions.

### **Experiment**

In our study, we mainly use two different family of models to conduct experiments, GPT-4o and LLaMA3.1 with 8B version (Dubey et.al., 2024). To set up the experiment, for each model, we prompted a fixed instruction of this task and incrementally added each word into context. We also highlight the current word in the prompt. After each word, the models were prompted with the question: "Is this word a transition between two events, marking an event boundary? Answer 'Yes' or 'No'." We not only documented the model's direct responses, but also probed the logits of 'Yes' and 'No' from the model outputs, which can be used to align with human density data later.

To mimic human-like event segmentation mechanisms, we suppose that humans may not use global memory to do this task. That is said, humans may have a relatively short memory when deciding before or after the word is two different events. Given this, we truncated original contexts

from beginning based on different context lengths, which means we only provide the most recent context of assigned context lengths as the story goes on. We selected six different context lengths: 64, 128, 256, 512, 1024 and unlimited tokens. By setting this up, we will be able to observe which context length may best align with human judgements.

### **Alignment Evaluation**

To evaluate the alignment between model and human response, we first need to normalize the human density data to the story length, at word level, as suggested by previous literatures (Kumar et.al., 2023). This will make two sets of data comparable. After normalization, we allowed a maximum lag of 50 words in assessing the alignment. The lag means human density data may be shifted forward or backward at maximum 50 words before conducting correlation, which takes into consideration of noises in the human responses (e.g., a delay in pressing the button). For each alignment measurement, we used the likelihood of answering ‘Yes’ from models to predict the normalized human density data. We used maximum beta weights of linear models within the lags to measure the alignment between model and human responses.

To consider prior preferences that humans may exhibit to specific words, we also added word frequency norms in the linear models. Specifically, these word frequency norms consist of the log of word frequencies for each word, and the log of context sensitivity. These two metrics measures how accessible are the words to human daily life experiences. We get these values from **SUBTLEXUS**, a public dataset based on subtle in movies and TV shows in the U.S. We also add these two variables in the linear models above to see whether control of word frequency norms help model responses (likelihood of ‘Yes’) to predict normalized human density data.

## Results

### Biased model responses over time in unlimited context

In our experiment, we first try out not limiting the context length of incremental prompting to the LLMs, which imitates human-like experimental setting. As we normalized both model responses (likelihood of ‘Yes’) and human density data, we found a consistent systematic biases from model responses in LLaMA-3.1-8B. Specifically, the overall likelihood of ‘Yes’ is increasing over time in all three stories, indicating that the models are more likely and often to segment the stories in their later parts, which deviates from human responses a lot (**Fig. 1**). In a real human-like responses, this should not increase over time since the event boundary is compared with two adjacent events not with the first event in the story. The model’s behavior in this setting might occur because of its biases in long-context learning. In transformer structure, the information in the first and last of the context typically get more attention in the computations, which is also known as **positional bias** (Vaswani et.al., 2017; Liu et.al., 2024). To overcome this bias and achieve more human-like responses, we limit the context length at 64, 128, 256, 512 and 1024 to see whether the alignment could be improved.



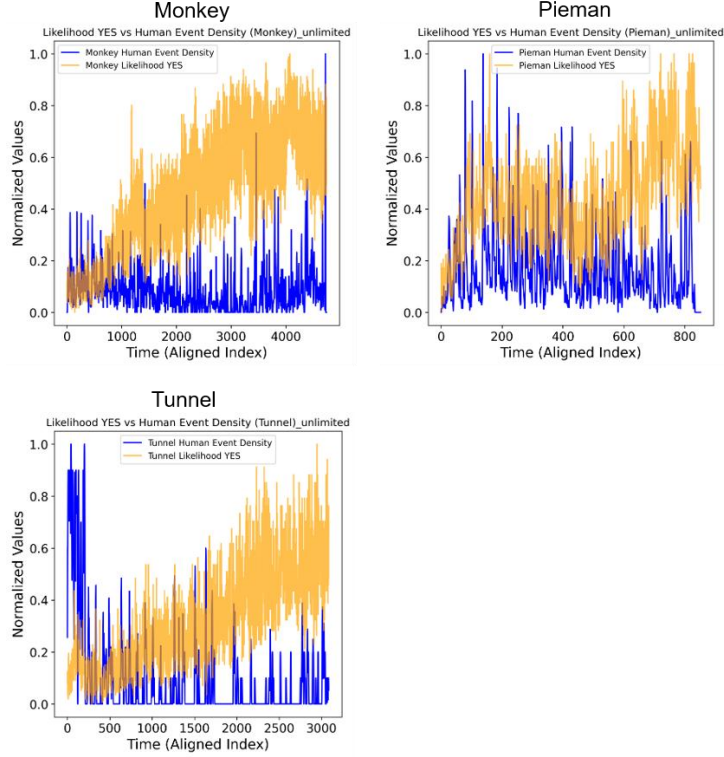


Figure 1: LLaMA-3.1-8B exhibits a systematic ‘lost in the middle’ bias in unlimited context setup.

### Shorter context lengths exhibit better alignment with human density data

To figure out the alignment between the model responses and human density data, while considering noises in human responses, we allow a maximum lag of 50 words (see methods) before conducting correlation analysis. We use the maximum correlation coefficient found in this range as the alignment between model responses and the human density data. As shown in **Fig. 2**, GPT-4o generally exhibits higher maximum correlation coefficient than LLaMA-3.1-8B model under the same setting, indicating a larger model may be helpful to perform this task in a more human-like manner. More importantly, we found that as we reduce the context length, the model typically exhibits better alignment with human density data. This suggests that as we enforce the model only

accessible to most recent information, their judgments are more human-like, which shows a memory mechanism in humans during the task.

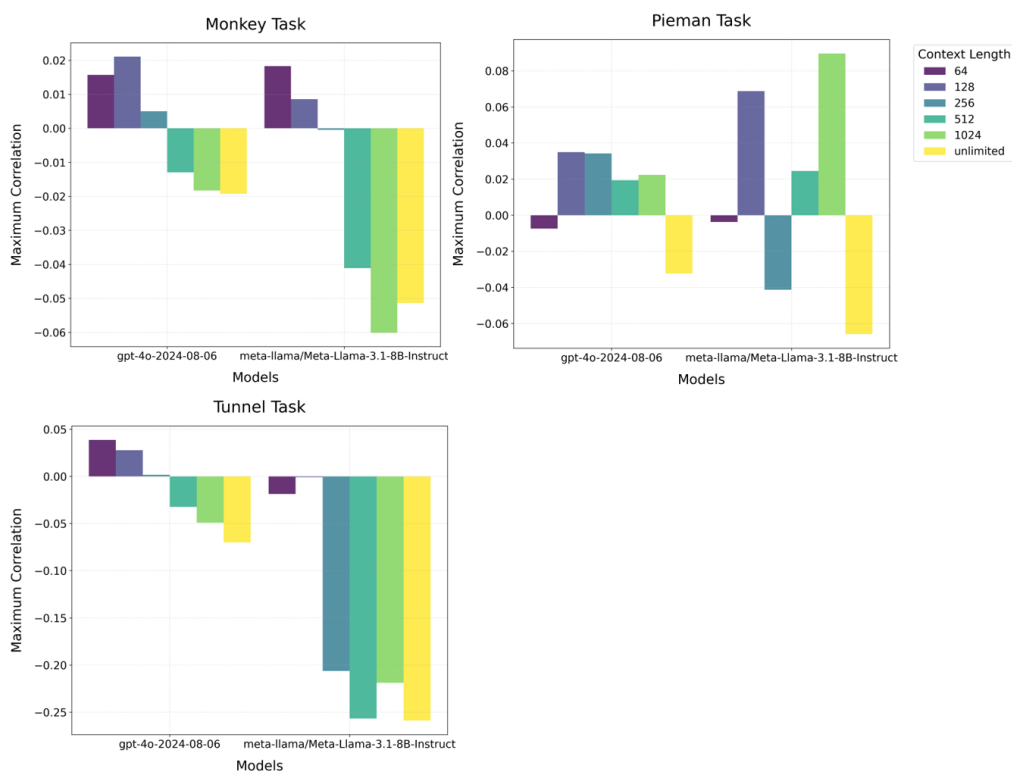


Figure 2: GPT-4o exhibits better alignment under the same setting than LLaMA-3.1-8B. Shorter context lengths exhibit better alignment with human density data.

### Word frequency norms slightly improve the alignment

As humans may perform the task not completely by semantic understanding of the overall story context, they may have some prior biases in specific kind of words. There might be two different factors affecting the judgement. First, if a word is not commonly accessible in human daily life experiences, hearing such word may bring special attention to humans and enhance the likelihood of response. Second, some words are not semantically meaningful, but may indicate a follow-up information about event transition (e.g., ‘but’, ‘where’ and ‘when’). This kind of words always indicate a context transition and may directly attract human participants to respond. These two

prior experiences can influence human’s judgements even without knowing what exactly the relevant event is. To consider this factor, we used two variables from **SUBTLEXUS** (see methods), log of frequency norms and context sensitivity to measure two factors mentioned above respectively.

We did a similar analysis as before, but adding these two variables as controlled variables in correlations. We still investigate the maximum correlation coefficient of model responses within the maximum lag of 50 words. The result suggests that adding those controlled variables slightly improves the alignment (**Fig. 3**). This result may suggest that in the pretraining of LLMs, these factors have already be taken into considered since they can represent on the distribution of training dataset. Therefore, adding these two factors does not provide additional information for the model responses.

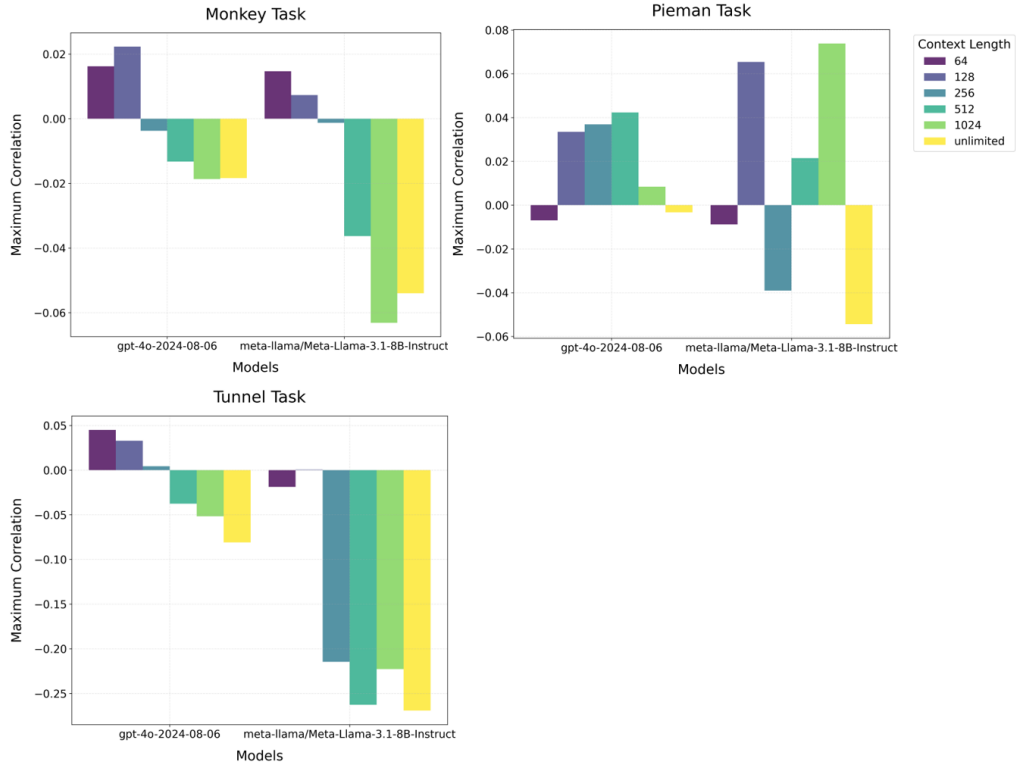


Figure 3: Adding two controlled variables slightly improve the performance of alignment of models.

## Discussion

Human event segmentation is crucial in daily life. In spite of classic theories like EST, how humans perceive events boundaries can be challenging to model in general and real-life like contexts. In our study, we used large language models to simulate human event segmentation behaviors, and tried to pin out the underlying cognitive insights about human event segmentation behaviors.

We used incremental word processing to LLM, trying to replicate how humans experience in the experiment. However, as we provided full context of the whole story, we found a systematic ‘lost in the middle’ bias by the language models. This may result from positional biases in transformers structures, which is indeed different from human word processing architecture. This may imply that the fundamental architecture of humans and language models should be taken into careful consideration in the comparison and analogy.

By fixing this fallback, we provided story context with varying limited context lengths. We found that models with shorter context lengths can yield better alignment with human density data. This result suggest that only very limited local information is utilized during human event boundary judgements. This setting may simulate human-like memory mechanism during the task that only adjacent information is memorized and used to judge. Finally, after we considered and controlled human prior experiences indicated by word frequency norms and context sensitivity in word processing, we found that it has very limited improvement on the alignment. This may suggest that the pretraining of language models already capture such features.

Although, we tried this experiment in the best of real-like experimental settings, there are a few limitations in our study. First, for the best of the result, our maximum correlation is relatively weak, especially compared with previous studies (Kumar et.al., 2023). However, in the original paper, they merged three stories into once. Their correlation is between predicted outputs and human

density data, where the predicted outputs is the result from the fitted linear model between Bayesian Surprise and human density data. The result is trivial since it ignores the direction of true correlation and only intends to fit the data into model. Therefore, we don't think this comparison is fair enough and informative. The weak correlation may indicate the noise of the human dataset or non-adaptive context length for different stories. In the future, we will need more tasks and data, as well as detailed investigations to optimize the performance. Second, the underlying representation in LLM about event boundary is poorly understood. By analyzing LLM's latent embeddings, combining with human density data may be helpful to approximate human mental representation of event boundary. Future research could potentially consider using Sparse Auto Encoders (SAE) to separate different events in one story, as well as figure out what tokens contribute most to an event segmentation judgment by dictionary learning (Trenton et.al., 2023). These interpretability approach will enable us to get closer to human minds by going through the language model's computations.

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## References

- Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6), e2218523120.
- Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., ... & Ganapathy, R. (2024). The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Kumar, M., Goldstein, A., Michelmann, S., Zacks, J. M., Hasson, U., & Norman, K. A. (2023). Bayesian surprise predicts human event segmentation in story listening. *Cognitive science*, 47(10), e13343.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157-173.
- Michelmann, S., Kumar, M., Norman, K. A., & Toneva, M. (2023). Large language models can segment narrative events similarly to humans. *arXiv preprint arXiv:2301.10297*.
- Radvansky, G. A., & Zacks, J. M. (2011). Event perception. *Wiley Interdisciplinary Reviews: Cognitive Science*, 2(6), 608-620.
- Reynolds, J. R., Zacks, J. M., & Braver, T. S. (2007). A computational model of event segmentation from perceptual prediction. *Cognitive science*, 31(4), 613-643.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah (2023). Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. *Transformer Circuits Thread*.
- Vaswani, A. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*.